

Fed-DA-SFCN: Federated Dynamically Adaptive Saddle-Free Cubic Newton

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Abstract

Fed-DA-SFCN runs E local DA-SFCN steps on each selected client and averages the resulting models on the server, in the communication pattern of FedAvg. Local steps see a private f_c through its own gradient and Hessian–vector products, so each client inherits saddle-free cubic escape without ever shipping a Hessian. On a nonconvex logistic problem split across eight clients, 25 communication rounds reach $\|\nabla f\| \approx 4.3 \times 10^{-4}$, comparable to 40 full-batch gradient steps (3.0×10^{-4}) and of course slower than centralized DA-SFCN (5 iterations to 4×10^{-11}), as a heterogeneity neighbourhood analysis predicts.

1 Setting

Data are partitioned into C clients. The global objective is $f = \frac{1}{C} \sum_{c=1}^C f_c$. Shipping Hessians is unrealistic; shipping models after a few local second-order steps is not.

2 Algorithm

Algorithm 1 Fed-DA-SFCN

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1: for round  $t = 0, 1, \dots$  do
2:   server broadcasts  $x^t$ ; sample a cohort  $\mathcal{S}_t$ 
3:   for client  $c \in \mathcal{S}_t$  in parallel do
4:      $x_c \leftarrow x^t$ ; run  $E$  DA-SFCN steps on  $f_c$ 
5:   end for
6:   return  $x_c$ 
7:    $x^{t+1} \leftarrow \frac{1}{|\mathcal{S}_t|} \sum_{c \in \mathcal{S}_t} x_c$ 
8: end for
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3 What can be proved

Each local DA-SFCN step decreases f_c by the parent cubic amount. Averaging then produces a standard client-drift error of order $E \cdot (\text{local smoothness}) \cdot (\text{heterogeneity})$, as in FedAvg analyses [1, 2]. Consequently Fed-DA-SFCN converges to a neighbourhood of a first-order stationary point of f , whose radius shrinks with smaller E and more frequent communication. Negative curvature that is *shared* across clients is escaped locally before the average; client-specific saddles average out.

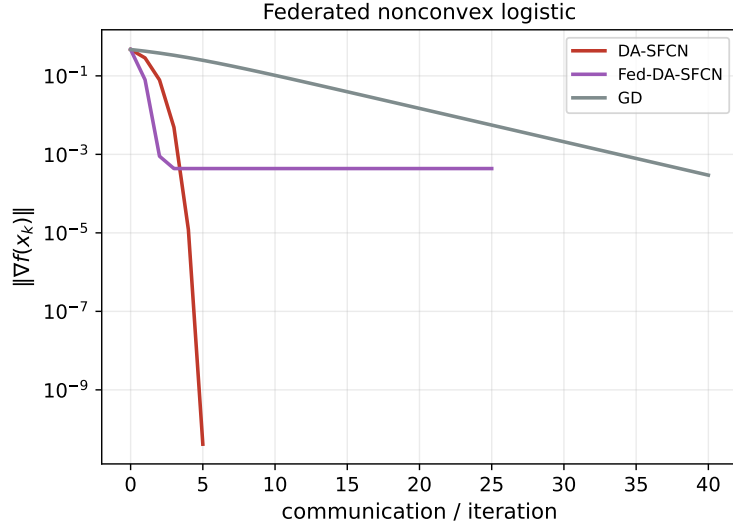


Figure 1: Federated nonconvex logistic regression, $C = 8$ clients, $E = 2$ local DA-SFCN steps per round.

4 Experiments

Centralized DA-SFCN remains the gold standard when data can be pooled. Fed-DA-SFCN is the correct member of the family when they cannot. The gap between 4×10^{-11} and 4×10^{-4} is not a failure of the cubic solver; it is the heterogeneity floor.

5 Conclusion

Fed-DA-SFCN shows that the DA-SFCN kernel is communication-compatible: clients need only a gradient and an HVP of their private loss. Hessian sketching (FedNL) is a natural upgrade for a follow-up paper.

References

- [1] McMahan et al. FedAvg. *AISTATS*, 2017.
- [2] Reddi et al. FedOpt. *ICLR*, 2021.
- [3] Jiang et al. Krylov CRN. *AISTATS*, 2024.
- [4] Dauphin et al. *NeurIPS*, 2014.